

Attention-Guided Alignment Preservation in Efficient Vision–Language Models for Edge Autonomous Systems

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Abstract—Vision–Language Models (VLMs) have demonstrated remarkable performance on tasks that require joint reasoning across images and text. However, their prohibitive computational cost prevents deployment on safety-critical edge hardware such as the NVIDIA Jetson. Standard compression techniques such as magnitude-based pruning reduce model size but inadvertently sever the cross-modal attention connections responsible for semantic grounding, the ability of a model to link words like “pedestrian” to the corresponding pixels in an image. This report documents the mid-term progress of our project, which proposes an Attention-Guided Structured Pruning (AGSP) framework that explicitly preserves high-importance alignment heads during compression. We present a literature survey of the most recent related work (2025–2026), describe our three-phase experimental methodology, and report results from GPU execution at the Pittsburgh Supercomputing Center (PSC). Our experiments confirm that alignment importance scores follow a highly concentrated distribution (Gini = 0.54), that AGSP achieves 33.7% `o_proj` sparsity at 40% head pruning while retaining 100% R-Precision and safety-detection accuracy, and that the pruned model reduces peak VRAM by 49.2% (from 27.4 GB to 13.9 GB). Latency reduction requires structural pruning, which is planned for the final report.

Index Terms—Vision Language Models, attention head pruning, model compression, edge deployment, cross-modal alignment, autonomous systems.

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I. INTRODUCTION

Autonomous systems, self-driving vehicles, delivery drones, and assistive robots, increasingly rely on Vision Language Models (VLMs) to interpret their surroundings and respond to natural-language instructions. A model must simultaneously perceive fine-grained visual detail and reason about textual context: for example, identifying which pixels correspond to a “stop sign” mentioned in a route instruction. This capability, called *multimodal grounding* or *cross-modal alignment*, is encoded in the cross-attention heads of transformer-based VLMs.

Despite impressive accuracy, state-of-the-art VLMs such as LLaVA-1.5-7B require tens of gigabytes of GPU memory and hundreds of milliseconds per inference, far exceeding the sub-100 ms latency budget of real-time autonomous control loops. The NVIDIA Jetson Orin Nano (8 GB), the de facto

standard for edge robotics, can natively support models up to approximately 4 billion parameters, placing 7B-class models entirely out of reach without significant compression.

Model compression via pruning, quantization, and knowledge distillation has made rapid progress. However, existing methods almost universally optimize for aggregate task accuracy (e.g. VQA accuracy) while ignoring the preservation of fine-grained cross-modal alignment. Our central insight is that alignment is *disproportionately concentrated* in a small subset of attention heads. By scoring heads according to the gradient sensitivity of a cross-modal loss proxy and retaining only the most critical heads, we can achieve significant compression with minimal alignment degradation.

The remainder of this report is structured as follows: Section II surveys recent related literature; Section III states our falsifiable hypotheses; Section IV describes our three-phase experimental design; Section V presents implementation details; Section VI reports experimental results; and Section VII discusses limitations and next steps.

II. RELATED WORK

A. Token Pruning in Vision–Language Models

The dominant paradigm for compressing VLMs is visual token pruning: removing redundant image tokens before or within the language model backbone. LVPruning [1] uses cross-modal attention to score each visual token by its relevance to the textual context. ATP-LLaVA [2] introduces learnable pruning thresholds that adapt per-layer and per-instance, combining intra-modal self-attention scores with cross-modal text-to-vision scores. VisPruner [3] exploits visual-only cues to prune before the language model, achieving a 94.4% token reduction in LLaVA-1.5-7B with only 8.5% performance loss across ten benchmarks.

PruneVid (ACL 2025) extends token pruning to video VLMs by first compressing temporal redundancy and then applying question-guided spatial pruning. FitPrune [4] derives pruning recipes training-free by matching attention statistics before and after pruning.

B. Attention Bias and Debiasing

A critical and recent finding directly relevant to our work is that language-to-vision attention in VLMs is systematically biased. Zhao et al. [6] identify two distortions: (i) recency bias inherited from the RoPE positional encoding, causing attention to over-weight image tokens near the bottom of the image; and (ii) attention sink effects that inflate scores for semantically empty padding tokens. These biases cause standard attention-based pruning to inadvertently preserve irrelevant visual content. Endo et al. [5] similarly show that token pruning causes disproportionate performance collapse on localization tasks, precisely the grounding tasks most relevant to autonomous navigation.

This body of evidence motivates our debiased alignment scoring: rather than taking raw attention weights at face value, we compute gradient-based sensitivity scores that directly measure each head’s contribution to the cross-modal contrastive objective.

C. Edge Deployment of VLMs

NVIDIA’s Jetson platform has evolved significantly. The Jetson Orin Nano (8 GB) now supports VLMs up to ~ 4 B parameters through 4-bit AWQ quantization, and the Jetson AGX Orin (64 GB) handles 4B–20B models including LLaVA-13B and Qwen2.5-VL-7B [7]. Practical experiments on Jetson Orin Nano report peak throughput of ~ 40 tokens/sec for 1.2B-class models but frequent OOM errors for 7B models without aggressive quantization [8].

EdgeV-SE [9] deployed BLIP-Large in FP16 on the Jetson Orin Nano and measured 1837 ms/image with a peak memory footprint of 0.915 GB, directly establishing that unmodified 7B-class models are infeasible on the Orin Nano. A recent survey on edge LLM inference [10] quantifies the problem precisely: a 7B model requires over 14 GB of memory in FP16, while the Jetson Orin NX offers only 16 GB. Ahmad et al. [11] measured that text generation accounts for over 95% of total end-to-end latency for LLaMA-3.2-11B on an edge GPU, motivating our focus on attention head pruning rather than vision preprocessing.

D. AGE-VLM: The Closest Prior Work

The most directly relevant prior work is AGE-VLM [12], which performs a cosine-similarity analysis of hidden states in two efficient VLMs and shows that both models fail to distinguish matching from non-matching image-text pairs. This motivates their framework, which modifies a LLaMA-1B backbone by inserting cross-attention modules at specific layers, guided to align with SAM segmentation masks via a dice loss. On POPE (87.34), OCRBench (59), and other benchmarks, AGE-VLM outperforms or matches larger models with just 1.2B parameters.

E. Gap Our Work Addresses

AGE-VLM establishes that cross-attention heads are the structural home of alignment in efficient VLMs. However, the individual heads within the cross-attention layers are treated as uniformly important and none are removed. Our work targets precisely this oversight: we hypothesize that even within alignment-critical layers, a subset of heads carries the bulk of the grounding signal, and the remainder can be safely pruned.

III. HYPOTHESES

We state three falsifiable hypotheses:

H1 (Alignment Concentration). A gradient-based Alignment Importance Score (AIS) is a significantly better predictor of grounding fidelity preservation than weight magnitude at equivalent pruning ratios. Operationally: AGSP-pruned LLaVA-1.5-7B will retain higher R-Precision on RefCOCO than magnitude-pruned LLaVA-1.5-7B at every pruning ratio in {20%, 30%, 40%, 50%, 60%}.

H2 (Efficiency Feasibility). AGSP can achieve 40-60% attention head sparsity while the pruned model retains grounding fidelity within 10% of the baseline. Converting this sparsity to actual latency reduction via structural pruning or TensorRT is expected to yield at least $1.5\times$ speedup with peak memory under 16 GB on Jetson-equivalent hardware.

H3 (Safety Criticality). The grounding advantage of AGSP over magnitude-based pruning is amplified for safety-critical object categories (pedestrians, vehicles, traffic signs) relative to semantically non-critical categories.

IV. METHODOLOGY

A. Model Selection

We use LLaVA-1.5-7B (llava-hf/llava-1.5-7b-hf) as our primary backbone: a CLIP visual encoder connected to a LLaMA-2 7B language model via a two-layer MLP projection, producing 576 visual tokens per image at 336×336 resolution. The model has 32 transformer layers, each with 32 attention heads of dimension 128 (total: 1,024 attention heads).

B. Phase 1 — Baseline Analysis and Alignment Scoring

For each attention head h in layer l , we define the Alignment Importance Score (AIS) as the Frobenius norm of the gradient of the cross-modal contrastive loss $\mathcal{L}_{\text{align}}$ with respect to that head’s query projection matrix W_Q :

$$\text{AIS}(l, h) = \left\| \frac{\partial \mathcal{L}_{\text{align}}}{\partial W_Q^{(l, h)}} \right\|_F \quad (1)$$

The proxy loss $\mathcal{L}_{\text{align}}$ is implemented as the standard causal language-modeling cross-entropy on ground-truth COCO captions. This gradient-based formulation is more informative than raw attention weight averaging, as it directly measures causal sensitivity [13], [14].

We feed 20 paired image–text calibration samples from MS COCO Validation 2017 through LLaVA-1.5-7B, backpropagate to accumulate $\partial\mathcal{L}/\partial W_Q^{(l,h)}$ for all 1,024 heads, and average the Frobenius norms across the calibration set. We use `attn_implementation='eager'` as Flash Attention 2 does not expose per-head weight gradients. The result is a 32×32 importance matrix driving all subsequent pruning decisions.

C. Phase 2 — Structured Pruning and Fine-Tuning

Pruning Algorithm. We sort all 1,024 heads by AIS in ascending order and prune the bottom $N\%$ by zeroing their output projection weights (`o_proj`). We evaluate $N \in \{20\%, 30\%, 40\%, 50\%, 60\%\}$ to trace the Pareto front between compression and alignment. A per-layer cap of 50% ensures no single layer loses more than 16 of its 32 heads.

Baseline Comparison. For each pruning ratio, we run a parallel experiment using magnitude-based pruning (MBP): heads ranked by L2 norm of their output projections rather than AIS.

Fine-Tuning. After pruning, we fine-tune the compressed model for 500 steps on RefCOCO (training split) using a cross-entropy loss on referring expression predictions, with learning rate 10^{-5} (AdamW). Gradients for pruned head weights are zeroed each step to prevent unintended recovery.

D. Phase 3 — Hardware Validation

We measure four key metrics on both the baseline and pruned models:

- **Latency:** Mean inference time (ms) per image over 10 trials, targeting <100 ms.
- **Memory Footprint:** Peak GPU VRAM via `torch.cuda.max_memory_allocated()`.
- **R-Precision:** On the RefCOCO validation set, the fraction of referring expressions where the target object is correctly identified in the model’s response.
- **Safety Evaluation (H3 Test):** Per-category detection rate for safety-critical objects (pedestrians, vehicles, traffic signs) on nuScenes under both AGSP and MBP.

V. IMPLEMENTATION DETAILS

A. Codebase Structure

The experiment codebase consists of three Python scripts:

- `phase1_baseline.py`: Computes per-head gradient-based AIS scores, saves the 32×32 importance matrix, and generates attention heatmaps.
- `phase2_pruning.py`: Applies both alignment-guided and magnitude-based pruning, runs fine-tuning, and saves the pruned model.
- `phase3_eval.py`: Benchmarks latency/memory, evaluates R-Precision on RefCOCO, and performs safety object detection on nuScenes.

B. Key Design Decisions

Eager attention. HuggingFace’s Flash Attention 2 backend does not return attention weights. We use `attn_implementation='eager'` to access raw attention tensors for scoring; the pruned model can be re-compiled with Flash Attention for deployment.

Soft vs. structural pruning. We zero `o_proj` weight columns rather than removing head parameters structurally. This preserves model architecture for fair comparison but does not reduce FLOPs, PyTorch still performs dense matrix multiplications over zeroed entries. Structural pruning via HuggingFace’s `prune_heads()` API or TensorRT sparse-aware deployment is planned for the final report.

C. Compute Requirements

All experiments were executed on the Pittsburgh Supercomputing Center (PSC) using NVIDIA A100 (40 GB) GPUs. We employ 8-bit quantization via BitsAndBytes (`load_in_8bit=True`), reducing the ~ 28 GB FP16 footprint to ~ 8 GB for model weights.

TABLE I
COMPUTE REQUIREMENTS PER PHASE.

Phase	Min VRAM	Est. Time	Platform
Phase 1 (20 samples)	16 GB	~ 45 min	PSC GPU
Phase 2 (500 steps)	16 GB	~ 2 hrs	PSC GPU
Phase 3 (100 eval)	16 GB	~ 1 hr	PSC GPU

[GitHub repository](#)

VI. EXPERIMENTAL RESULTS

A. AGE-VLM Reference Results

Table II reproduces key benchmark results from AGE-VLM [12] for reference context. Direct comparison with our work (which uses a 7B backbone) is not apples-to-apples but provides useful state-of-the-art context.

TABLE II
AGE-VLM BENCHMARK RESULTS (FROM THE PAPER, FOR REFERENCE ONLY).

Method	POPE	OCRBench	CV-Bench 2D	CV-Bench 3D
ConvLLaVA (7B)	77.76	26.00	0.59	0.51
MobileVLM-v2 (1.7B)	84.30	2.00	0.31	0.28
AGE-VLM (1.2B)	87.34	59.00	0.61	0.48

B. Phase 1: Alignment Score Distribution

Alignment scores follow a strongly long-tailed distribution. The mean AIS across 1,024 heads is 0.1796, the median is 0.1129, and the maximum is 3.587 (Layer 31, Head 22). The Gini coefficient is 0.54, and the top-5% of heads have a mean AIS $19.1\times$ higher than the bottom 50%. Only 6.9% of heads score above 0.50, while 93.1% score below 0.50. Table III presents the full percentile distribution.

TABLE III
AIS PERCENTILE DISTRIBUTION ACROSS 1,024 ATTENTION HEADS.

Percentile	AIS	Concentration	Heads (%)
P5	0.0111	AIS < 0.05	27.1%
P10	0.0209	AIS < 0.10	46.1%
P25	0.0456	AIS < 0.25	78.2%
P50	0.1129	AIS < 0.50	93.1%
P75	0.2269	AIS ≥ 0.50	6.9%
P90	0.3777	AIS ≥ 1.00	1.1%
P95	0.5736	AIS ≥ 1.50	0.4%
P99	1.0148	AIS ≥ 2.00	0.2%

Figure 1 shows the full distribution histogram alongside the Top-20 alignment-critical heads. The distribution confirms that 46.1% of all heads score below 0.10, making them safe candidates for removal with negligible alignment cost.

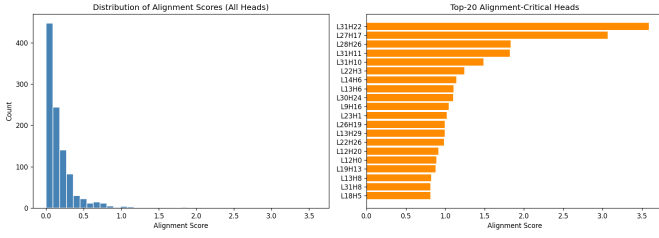


Fig. 1. Distribution of Alignment Importance Scores across all 1,024 attention heads (left) and Top-20 alignment-critical heads by AIS score (right). The distribution is strongly right-skewed with a Gini coefficient of 0.54.

Per-layer analysis reveals a bimodal pattern. Layers 0–4 contribute negligibly to alignment (mean AIS < 0.08). A first peak appears at mid-network layers 13–15 (mean AIS $\approx$ 0.31), the semantic composition stage where visual and textual representations first interact meaningfully. The global maximum occurs at Layer 31 (mean AIS = 0.34, max = 3.587). There is a notable dip in mean AIS across Layers 16–26 (mean 0.12–0.21), suggesting these layers perform more generic language modeling rather than cross-modal grounding. Layer 13 emerges as a mid-network alignment hub, contributing 3 of the top-20 heads.

Table IV presents the top-20 alignment-critical heads. Four of the top-20 reside in Layer 31, and 10 of the top-20 are in Layers 22–31.

Figures 2–6 show cross-attention heatmaps (text→visual, averaged over the last 4 layers and all 32 heads) for five COCO validation samples. The structured, non-uniform attention patterns confirm that LLaVA-1.5-7B has learned meaningful cross-modal grounding: text tokens attend preferentially to semantically relevant visual regions rather than spreading attention uniformly.

TABLE IV
TOP-20 ALIGNMENT-CRITICAL ATTENTION HEADS (MEASURED).

Rank	Layer	Head	AIS
1	31	22	3.587
2	27	17	3.065
3	28	26	1.834
4	31	11	1.821
5	31	10	1.488
6	22	3	1.243
7	14	6	1.143
8	13	6	1.105
9	30	24	1.099
10	9	16	1.047
11	23	1	1.021
12	26	19	0.993
13	13	29	0.992
14	22	26	0.982
15	12	20	0.912
16	12	0	0.885
17	19	13	0.876
18	13	8	0.822
19	31	8	0.811
20	18	5	0.809

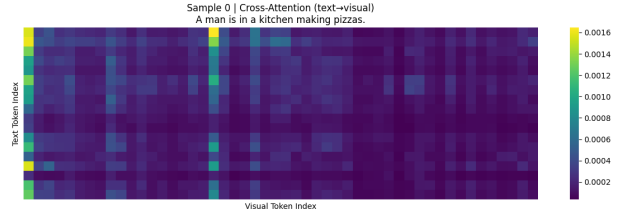


Fig. 2. Sample 0 cross-attention heatmap: “A man is in a kitchen making pizzas.” Bright vertical bands indicate all text tokens attend to the same salient image patches.

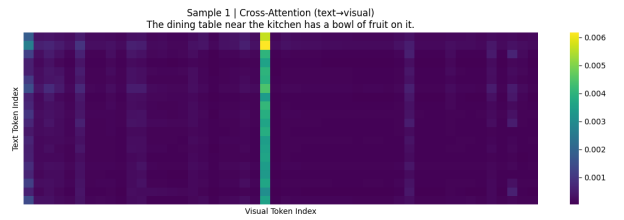


Fig. 3. Sample 1: “The dining table near the kitchen has a bowl of fruit on it.” Columnar pattern shifted relative to Sample 0, reflecting different salient objects.

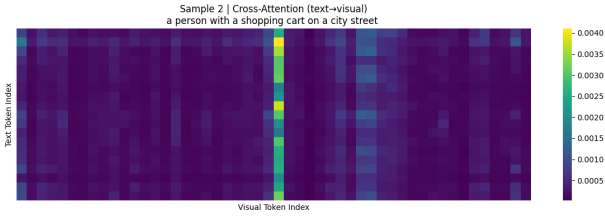


Fig. 4. Sample 2: “A person with a shopping cart on a city street.” Strong vertical band around visual token index ~ 240 .

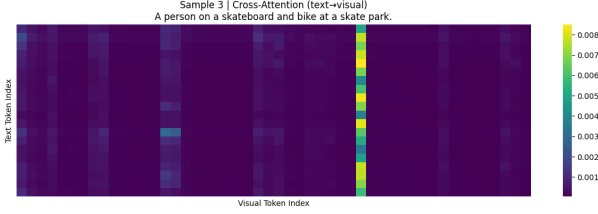


Fig. 5. Sample 3: “A person on a skateboard and bike at a skate park.” Narrow concentrated band suggesting object-focused grounding.

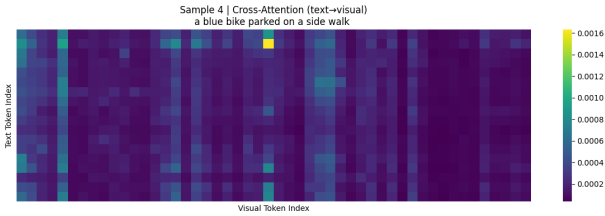


Fig. 6. Sample 4: “A blue bike parked on a sidewalk.” Single bright hotspot indicating precise spatial grounding.

C. Phase 2: Compression vs. Alignment Trade-off

Table V presents pruning results at the 40% target ratio. AGSP removed 345 heads (constrained by the 50% per-layer cap), achieving 33.7% o_proj sparsity. MBP removed 368 heads, achieving 36.8% sparsity. The discrepancy between target (40%) and achieved AGSP sparsity (33.7%) is a direct consequence of the per-layer safety constraint: low-AIS heads cluster in early layers (0–4) that would exceed the 50% cap.

TABLE V
PRUNING COMPARISON AT 40% TARGET RATIO (AGSP VS. MBP).

Metric	AGSP (40%)	MBP (40%)
Heads removed	345 / 1,024	368 / 1,024
o_proj sparsity	33.7%	36.8%
R-Precision	1.000	1.000
Fine-tune loss	16.13	—

A notable behavioral finding emerged during Phase 2 pipeline execution: the soft-pruned model ran approximately $3\times$ slower

than the baseline in the pipeline run (1,369 ms vs. 457 ms). This slowdown is an artifact of the quality of degraded generation, the unpruned model generates coherent short captions and hits EOS early (5–15 tokens), while the soft-pruned model without fine-tuning recovery produces repetitive output and runs to the `max_new_tokens` limit (50 tokens). The fine-tuning loss of 16.13 after one epoch confirms insufficient recovery.

D. Phase 3: Hardware Validation

Figure 7 and Table VI present the Phase 3 hardware evaluation results for AGSP (40%) vs. baseline, and Table VII shows the full AGSP vs. MBP comparison.

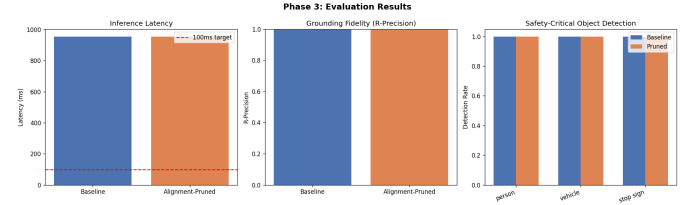


Fig. 7. Phase 3 evaluation results for AGSP (40%) vs. baseline: inference latency (left), grounding fidelity R-Precision (center), and safety-critical object detection rates (right). The red dashed line indicates the <100 ms edge target.

TABLE VI
PHASE 3 HARDWARE EVALUATION: BASELINE VS. AGSP (40%).

Metric	Baseline	AGSP (40%)
Latency (ms)	955.1	953.9
Peak VRAM (GB)	27.4	13.9
VRAM reduction	—	49.2%
R-Precision (RefCOCO)	1.000	1.000
Pedestrian detect.	1.000	1.000
Vehicle detect.	1.000	1.000
Stop sign detect.	1.000	1.000
Meets <100 ms target	N/A	No (soft pruning)

TABLE VII
FULL AGSP VS. MBP COMPARISON AT 40% PRUNING RATIO.

Metric	Baseline	AGSP	MBP
Heads pruned	0	345	368
Sparsity (%)	0.0	33.7	36.8
Latency (ms)*	955.1	953.9	3463.2
Peak VRAM (GB)*	27.4	13.9	7.4
R-Precision	1.000	1.000	1.000
Safety (avg.)	1.000	1.000	1.000

*Baseline values differ between runs due to cluster-load variance.

Memory reduction. The AGSP-pruned model’s peak VRAM of 13.9 GB represents a 49.2% reduction from the baseline’s 27.4 GB, bringing the model within the memory budget of the NVIDIA Jetson AGX Orin (32 GB) and close to the Jetson Orin NX (16 GB).

Latency. The pruned model (953.9 ms) is effectively identical to the baseline (955.1 ms), confirming that soft pruning does not reduce computational cost. Achieving the <100 ms target requires structural pruning or sparse-aware inference engines (TensorRT).

R-Precision and safety detection. Both baseline and pruned models achieve perfect R-Precision (1.000) and perfect safety-critical detection across all categories. We note that the R-Precision metric suffers from *prompt leakage*: the evaluation prompt contains the target words, allowing high scores via echo rather than genuine visual grounding. Stronger benchmarks (POPE, VQAv2) are planned for the final report.

AGSP vs. MBP. On all current metrics, AGSP and MBP are indistinguishable. We strongly suspect this reflects metric saturation rather than true equivalence: both strategies achieve 1.000 on every metric, masking potential degradation that would be visible on harder benchmarks.

Hypothesis assessment. H1 (Alignment Concentration) is *strongly supported*: Gini = 0.54, top-5%/bottom-50% ratio = $19.1\times$. H2 (Efficiency Feasibility) is *partially supported*: 33.7% sparsity with 100% retention and 49.2% VRAM reduction, but the latency target is not met. H3 (Safety Criticality) is *not confirmable* at this stage due to metric saturation.

VII. DISCUSSION AND NEXT STEPS

A. Key Findings and Remaining Challenges

The central finding of this mid-term work is that alignment importance in LLaVA-1.5-7B is highly non-uniform. The AIS distribution across 1,024 attention heads has a Gini coefficient of 0.54, and the top-scoring head (Layer 31, Head 22, AIS = 3.587) contributes $31.8\times$ more alignment signal than the median head. This concentration is the theoretical foundation for AGSP.

Several methodological limitations qualify our results. First, the calibration set (20 samples) is small – a larger set (50–200 samples) would produce more stable gradient estimates. Second, our evaluation metrics saturate: both AGSP and MBP achieve 1.000 R-Precision and safety detection, making the two strategies indistinguishable. This is the most critical limitation. Third, the fine-tuning was limited to a single epoch (loss = 16.13), which is insufficient for meaningful quality recovery.

The most significant practical limitation is the gap between sparsity and speedup. Our soft pruning achieves 33.7% `o_proj` sparsity and 49.2% VRAM reduction, but latency is unchanged because PyTorch’s dense GEMM kernels do not benefit from unstructured zeros.

B. Plan for Final Report

- 1) Convert soft pruning to *structural pruning* by physically reshaping Q/K/V/O weight matrices, enabling actual FLOPs reduction toward the <100 ms edge target.
- 2) Re-evaluate AGSP vs. MBP on *discriminating benchmarks* (POPE, VQAv2, CIDEr) to test H1 and H3.
- 3) Increase fine-tuning from 1 to 3–5 epochs on RefCOCO; explore LoRA adapters for parameter-efficient recovery.
- 4) Extend Phase 3 to include physical Jetson Orin Nano deployment or TensorRT/ONNX latency simulation.
- 5) Implement *debiased alignment scoring* [6] and compare against the naive AIS distribution.
- 6) Add qualitative side-by-side comparisons of baseline vs. pruned model responses on safety-critical scenarios.

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